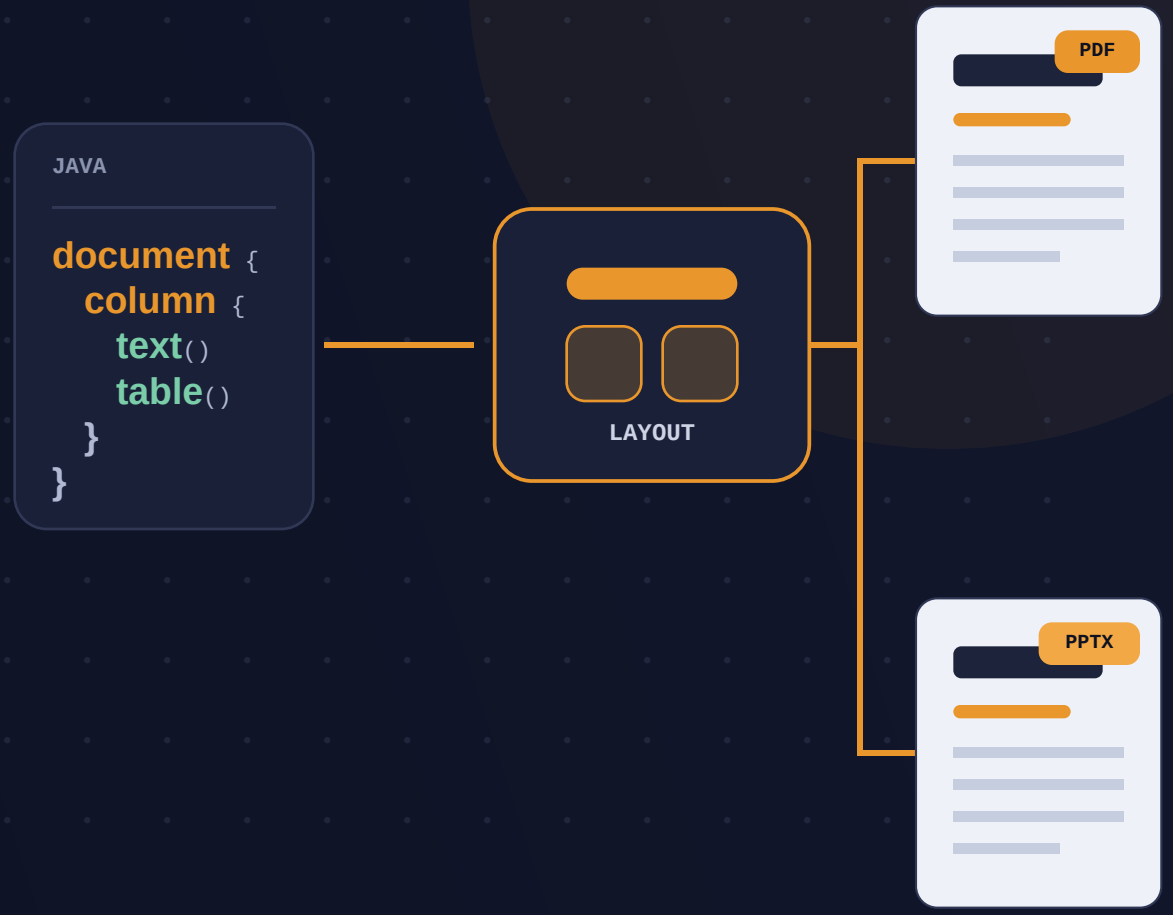
AVAILABLE ON MAVEN CENTRAL

GraphCompose

Declarative document generation for Java

```
io.github.demchaav:graph-compose
v2.1.1
```

JAVA 17+ · PDF · PPTX · AUTO-PAGINATION



One model. PDF and PPTX outputs.

Code › layout › finished document

Open-source Java library · MIT License

HOW IT WORKS

From one Java file to a finished document

You describe the document semantically — sections, rows, tables, charts, shapes, layers — and the engine resolves the rest. No manual coordinates, no XML templates, no per-format authoring.

1

AUTHOR

A fluent DSL describes intent, not geometry.

2

MEASURE

Every node measured twice, deterministically.

3

PAGINATE

The flow splits across pages row by row.

4

RENDER

A backend writes the bytes — PDF or PPTX.

Deterministic

The same input renders the same bytes on every machine. Layout is reproducible, not best-effort.

Regression-tested

layoutSnapshot() captures the resolved geometry, so a layout change fails a test instead of surfacing in a release.

One model, two formats

The same composition emits this PDF and an editable PPTX. Nothing is authored twice, so the two cannot drift.

MEASURED, NOT CLAIMED

GraphCompose against the field

The same documents through three engines — render time and peak heap, measured by this repository's own harness. Every figure on this page and the next is read from the result file at render time, not typed in.

Report size	GraphCompose	iText 9	JasperReports
1 page · 3 lines	2.0 ms · 0.2 MB	2.1 ms · 0.2 MB	4.2 ms · 0.2 MB
40 rows	4.1 ms · 1.0 MB	10.0 ms · 3.0 MB	8.3 ms · 2.5 MB
200 rows	10.9 ms · 4.0 MB	36.6 ms · 13.6 MB	13.5 ms · 10.8 MB
1000 rows	42.8 ms · 19.3 MB	189.1 ms · 67.3 MB	43.7 ms · 52.5 MB

4.4x
faster than iText 9 at 1000 rows

3.5x
lighter than iText 9

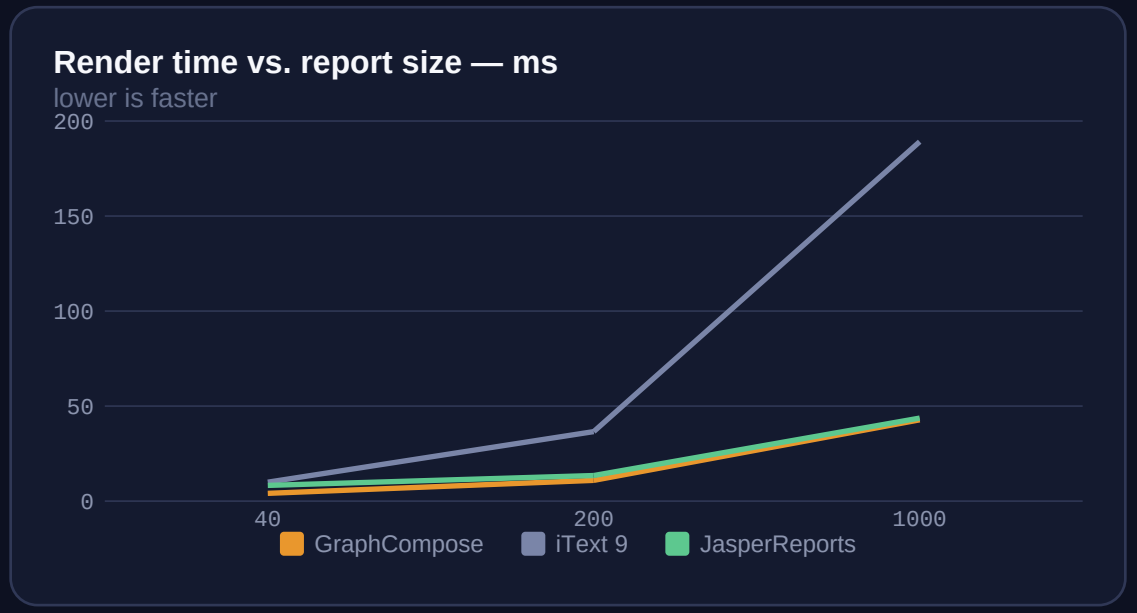
2.7x
lighter than JasperReports

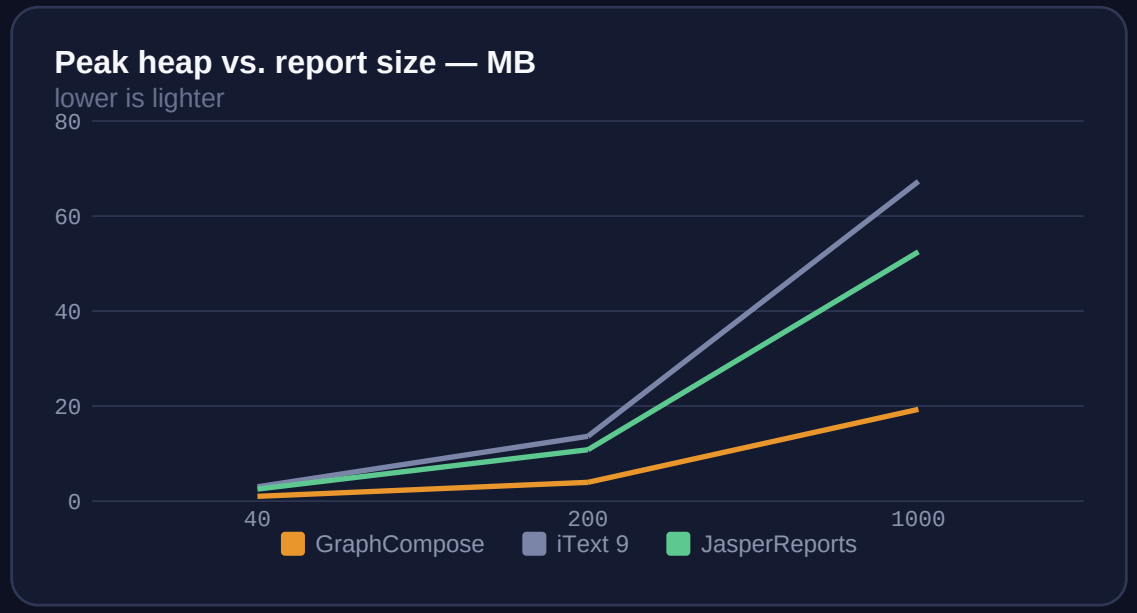
Measured 2026-08-04 22:16:04 · 50 warmup / 100 measurement iterations on one machine. Read the ratios rather than the milliseconds: all three engines are measured in the same run, so their proportions survive a change of hardware while the timings do not.

SCALING

What changes as the report grows

From 40 to 1000 rows the time lead over iText widens; JasperReports closes to roughly the same render time at the top size, while GraphCompose stays markedly lighter on memory than both throughout. Every series reads from the same file as the table on the previous page.





Both charts are native vector output: the engine compiles them to the same shapes, lines and text frames as everything else on the page — no image is embedded, and the numbers come from the file, not the caption.